9. Coupling

The lesson that explains why a Tesla coil is not a transformer

Two tuned circuits handing energy back and forth

Lesson 9 · about 75 minutes · everyone builds; battery power only

What you need

- Two identical pendulums — same length, same weight
- A horizontal string or dowel to hang them from
- Two identical LC tanks from Lesson 8
- Signal generator and oscilloscope, or the multimeter method
- A ruler, to set and measure coil spacing

Predict first

Two identical pendulums hang from the same slack string. You start one swinging and hold the other still, then let go. What happens to each over the next minute? Draw both, one under the other, against time.

Do it — the pendulums first

1. Hang two identical pendulums from a common horizontal string with some slack in it. Start one; leave the other at rest.
2. Watch for a full minute. The first slows and stops; the second, which nobody touched, is now swinging at full amplitude. Then it reverses, and the energy comes back. It keeps trading, indefinitely, until friction wins.
3. **Change the coupling.** Tighten the string. The trade happens faster. Slacken it right off and the second pendulum barely moves at all.
4. **Break the match.** Shorten one pendulum so the two no longer share a frequency. Now almost nothing transfers, at any coupling. *Matching matters more than coupling.*

Then the circuits

1. Set up two identical LC tanks, tuned to the same f_0, with their coils a few centimetres apart and coaxial.
2. Drive the first at f_0 and measure the voltage across the second.
3. Slide them closer. The transfer rises and the single sharp peak splits into *two* peaks either side of f_0 — an over-coupled pair no longer has one resonance, it has two.
4. Detune one tank with a different capacitor. Transfer collapses, exactly as with the shortened pendulum.

What it means

Coupling coefficient	Written k, from 0 (no shared field) to 1 (all of it shared). An iron-cored mains transformer runs at $k \approx 0.95$. A Tesla coil runs deliberately *loose*, around $k = 0.1$ to 0.2.
Why loose is better here	Energy takes many cycles to cross a loosely coupled pair, and during those cycles the secondary's voltage builds and builds. Couple them tightly and the transfer is fast but the secondary never gets the chance to ring up. Tight coupling also puts so much voltage across the secondary's own turns that it arcs internally and destroys itself.
Both must be tuned the same	This is what the pendulum experiment proves. Coupling with mismatched frequencies transfers almost nothing.
Energy goes both ways	The pendulums traded back and forth, and so do the coils. In a real coil we want the transfer stopped at the moment the secondary is fullest — which is exactly what the spark gap does in Lesson 10, by quenching.
This is why it is not a transformer	A transformer trades voltage for current by turns ratio alone. A Tesla coil does that *and* accumulates energy over many cycles of a shared resonance. That second mechanism is where most of the voltage comes from.

The whole machine, in one sentence you can now parse

The transformer charges the capacitor; the gap dumps it into the primary tank, which rings at f_0; the secondary is tuned to the same f_0 and loosely coupled, so over many cycles it takes that energy and rings up to a voltage far higher than the transformer alone could make; and the gap quenches at the right moment so the energy does not slosh back. Lessons 6, 7, 8 and 9 are those four clauses.

> **This lesson is low voltage.** Everything here runs from batteries or a small plug-in supply and cannot hurt you. That changes at Lesson 10. Build the habits now — one hand, discharge before touching, check before you power up — while the stakes are nothing, so that they are automatic when the stakes are real.